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Generalized multimodal depression severity prediction method based on topic prompts

Di Zhao [a], Qi Li [a,b,c,*], Qingyu Na [a], Hongyu Xie [a], Tingyun Jiang [d]

[a] School of Computer Science and Technology, Changchun University of Science and Technology, Changchun, 130022, China
[b] Jilin Provincial International Joint Research Center of Brain Informatics and Intelligence Science, Changchun, 130022, China
[c] Zhongshan Institute of Changchun University of Science and Technology, Zhongshan, 528437, China
[d] Department of Psychiatry, The Third People's Hospital of Zhongshan, Zhongshan, 528451, China

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ABSTRACT

Multimodal depression severity prediction holds significant potential for clinical diagnosis. However, existing methods tend to overfit in data-scarce environments and lack generalizability across diverse cultural contexts. To address these challenges, we propose a Topic Prompts Multimodal Network (TPMN), a novel framework that integrates topic-guided prompt learning with a Wide & Deep architecture for robust depression prediction. Specifically, topic prompts — contextual cues derived from underlying topic patterns in patient interviews — are constructed using a Latent Dirichlet Allocation (LDA)-based prompt collection, capturing personalized semantic topics to guide multimodal feature alignment. These prompts help the model prioritize clinically relevant interactions across modalities via a cross-attention coupling function, where complex transformations dynamically align multimodal features, effectively mitigating overfitting while enhancing interpretability. Extensive experiments on AVEC2013 and AVEC2014 demonstrate TPMN's superiority over state-of-the-art methods, achieving a 53% reduction in MAE (2.86 vs. SOTA 6.08) and 58% lower RMSE (3.36 vs. SOTA 7.96) on average. Notably, experiments on our self-collected Chinese dataset show that TPMN outperforms existing baseline methods, demonstrating its potential as a scalable diagnostic tool for diverse populations. Furthermore, zero-shot evaluation across cultural datasets demonstrates TPMN's strong generalizability, with results comparable to baseline methods. Beyond its technical contributions, this work bridges the gap between data-driven AI and clinical needs, offering a principled approach to personalized and culturally adaptive mental health assessment.

1. Introduction

Depression has become a common mental disorder. Clinical diagnosis of depression primarily relies on psychological assessment scale and doctor interviews, with the final diagnosis made subjectively by doctors based on their experience. This requires doctors to possess extensive clinical experience and superior diagnostic skills. However, there is a shortage of experienced doctors. Therefore, exploring simple, objective, and accurate auxiliary diagnostic methods for depression has long been a significant challenge [1,2].

In exploring objective depression recognition methods, multimodal approaches that integrate diverse data channels — such as text, audio, and video cues — have demonstrated superior accuracy compared to unimodal methods [3]. Recent advancements in multimodal depression detection leverage inputs such as conversational text, facial expressions, and vocal prosody, employing various fusion strategies to enhance diagnostic performance [4–6]. These strategies can be broadly categorized into feature fusion and decision fusion, each with its own strengths and limitations. For instance, M. A. Uddin et al. [7] proposed a Deep Multi-Modal Network that captures spatial and temporal features using multimodal factorized bilinear pooling (MFB), a feature fusion technique combining multimodal data at the feature level. Similarly, Huiting Fan et al. [6] introduced a Transformer-based feature-enhanced network for multimodal depression detection, which extracts features from video, audio, and rPPG modalities and employs a Graph Fusion Network for decision-level fusion, achieving competitive performance.

However, these multimodal fusion methods often generate high-dimensional feature representations, particularly from video data, requiring large-scale datasets for effective model training and generalization [8]. In practice, depression detection — like many other medical applications — faces significant challenges due to data scarcity [9]. This limitation exacerbates the risk of overfitting in traditional deep learning models, which may perform well on training data but fail to generalize

* Corresponding author at: School of Computer Science and Technology, Changchun University of Science and Technology, Changchun, 130022, China.
E-mail address: liqi@cust.edu.cn (Q. Li).

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to unseen samples [10]. Moreover, existing methods often overlook the cultural and linguistic diversity of patients, further limiting their applicability in real-world clinical settings [11].

To mitigate overfitting caused by data scarcity in depression detection tasks, pre-trained models have been identified as a reliable solution [12]. When fine-tuned on depression detection tasks, these pre-trained models can capitalize on the knowledge accrued during pretraining, enhancing their performance on new, unseen depression datasets [13,14]. However, the extensive size of these models and the limited training data can pose difficulties in fine-tuning the entire model, leading to a loss of useful knowledge acquired during the pretraining phase, a phenomenon referred to as catastrophic forgetting [12]. To address this challenge, using language prompt learning methods and provide a mechanism for prompt adaptation while maintaining the original weights frozen, reducing catastrophic forgetting [15]. However, it is sensitive to the selection of prompts, and exploring the best prompts and providing a mechanism to adapt to the prompt are key to solving the catastrophic forgetting and improving the model's ability to generalize to new datasets.

In actual diagnostic interviews, it is observed that doctors assess the interviewee's emotional state by examining facial activities and vocal cues in response to questions on various topics, thereby analyzing the emotional information conveyed in the interviewee's answers [3]. During the interview process, doctors focus on different topics based on individual information. For example, postpartum depressed patients may show a more obvious tendency towards depression when talking about childbirth, while depressed patients who have experienced childhood trauma may show a more obvious tendency towards depression when talking about their childhood experiences. Drawing on the diagnostic process of doctors, prompt-based topic modeling methods have been developed to achieve effective depression detection under conditions of limited annotated training samples [16,17]. However, prompt-based methods face challenges in personalized depression recognition: (1) At test time, the individual's specific depression-related information is unknown, making it unfeasible to train topic prompts that are unrelated to the individual's depression; (2) Even if individual prompts are available, knowledge sharing between similar individuals is hindered; (3) Although using a shared topic prompt across all individuals allows for knowledge sharing, the forgetting issue still persists.

Inspired by personalized recommendation systems, we adopt a personalized approach, as exemplified by Google's Wide & Deep model [18]. The Wide component of this model employs logistic regression to retain users' historical data information, whereas the Deep component utilizes a deep learning model to learn generalized features. In this paper, the Wide component of the model retains personal information, while the Deep component integrates a topic prompt pool. This approach integrates personal information with relevant topics from the prompt pool, enabling prompt knowledge sharing among similar individuals while preserving the independence of prompts in other cases. This helps address the challenge of prompt exploration in fine-tuning pre-trained models for depression detection.

The effective integration of prompts into the model is essential for their coordinated function with other model components. A dynamic interaction mechanism between modalities has been proposed to facilitate effective integration and utilization of prompt information across different modalities. This mechanism adjusted the weight and influence of prompts in different models, contingent upon the input data's characteristics and the model's prediction objectives. This approach not only enhances the model's adaptability to individual differences but also promotes a deeper fusion of information between disparate data modalities.

This study introduces TPMN, an innovative framework for predicting depression severity. It uses the Wide & Deep architecture and dynamic interaction mechanisms to incorporate personalized topic prompts into pre-trained large models. This approach enhances the performance and generalization of multimodal severity prediction, addressing individual variability and providing simple, objective, and accurate methods for depression severity prediction. This work aims to address two critical limitations in existing multimodal depression prediction methods: (1) overfitting in low-data scenarios due to excessive reliance on fine-tuning limited samples, and (2) poor cross-cultural generalizability resulting from neglecting culturally invariant clinical patterns. To overcome these challenges, we propose the Topic Prompts Multimodal Network (TPMN) with the following innovations:

1. Wide & Deep Architecture for Balanced Learning: By combining memorized personal traits (wide component) with deep hierarchical representations of multimodal data, TPMN effectively balances memorization and generalization, mitigating overfitting while preserving personalized insights.

2. Dynamic Prompt Coupling Mechanism: This mechanism dynamically integrates topic-aware prompts with modality-specific features, enabling the discovery of shared topic patterns among individuals with similar clinical profiles. By leveraging personalized topic prompts derived from latent semantic structures, TPMN captures common depression features that transcend cultural or linguistic differences, thereby enhancing cross-cultural generalizability.

2. Related work

2.1. Multimodal depression detection: Background and early developments

The field of multimodal depression detection has evolved significantly over the past decade, driven by advancements in machine learning and the increasing availability of multimodal data. Early studies primarily focused on unimodal approaches, leveraging either text, audio, or video cues to detect depressive symptoms. However, the limited expressiveness of single modalities often resulted in suboptimal performance, prompting researchers to explore multimodal fusion techniques.

One of the earliest multimodal frameworks for depression detection was proposed by Valstar et al. (2013) [19], who introduced the AVEC challenge series to benchmark multimodal methods using audio, video, and text data. Their work demonstrated that combining facial expressions, vocal prosody, and linguistic features could significantly improve depression recognition accuracy compared to unimodal approaches. This laid the foundation for subsequent research in multimodal fusion.

In 2013, Alghowinem et al. [20] proposed a multimodal framework that integrated audio and video features to detect depression. Their approach utilized handcrafted features, such as Mel-frequency cepstral coefficients (MFCC s) for audio and facial action units (AUs) for video, achieving promising results on small-scale datasets. However, the reliance on handcrafted features limited the generalizability of these methods.

The advent of deep learning in the late 2010s revolutionized multimodal depression detection. He et al. (2015) [21] proposed a deep multimodal network that automatically learned feature representations from raw audio, video, and text data. Their approach demonstrated superior performance over traditional methods, highlighting the potential of deep learning for multimodal fusion. Similarly, Gong et al. (2017) [17] introduced a topic modeling-based multimodal framework that combined text and audio features to detect depression, emphasizing the importance of semantic and acoustic cues in depression recognition.

Despite these advancements, early multimodal methods faced several challenges, including high-dimensional feature representations, data scarcity, and limited generalizability. These limitations motivated researchers to explore more sophisticated fusion strategies, such as attention mechanisms and transformer-based architectures, which have become dominant in recent years. For instance, in feature fusion, attention mechanisms have been widely adopted to dynamically weight the importance of different modalities, enabling the model to focus on

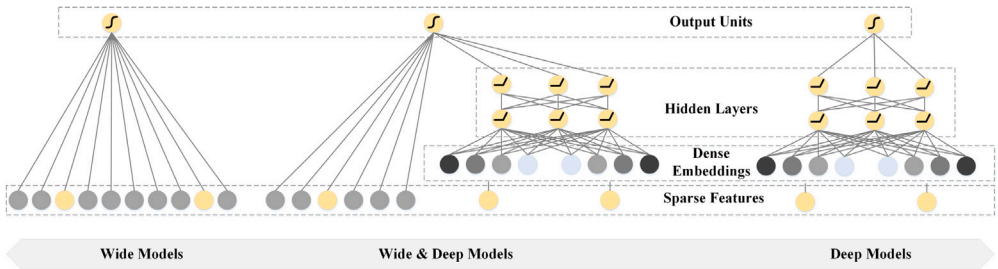

Fig. 1. Schematic diagram of Wide&Deep architecture.

the most relevant features for depression detection [6]. Similarly, in decision fusion, transformer-based architectures have been employed to capture long-range dependencies and contextual information across modalities, significantly improving the model's ability to integrate multimodal data [7].

2.2. Prompt learning

Prompt learning, originally introduced in GPT-3 [22], offers an alternative approach to reframing downstream tasks to resemble tasks already solved during pre-training. During fine-tuning, a model takes an input x and predicts an output y as $p(y|x)$. In the prompting process, the original input x is modified by a prompt template into a new string with unfilled slots, and the pre-trained language model fills in these slots to obtain the output y [23]. The advantage of prompt learning lies in its ability to adapt to tasks with minimal or even zero samples, by capitalizing on its strong transferability across tasks. This transferability arises from the language model's requirement to learn multiple implicit tasks on a large-scale pre-training corpus. However, this transferability requires substantial model capacity to handle potential implicit tasks and is highly sensitive to prompt formulation. This sensitivity introduces the challenge of prompt engineering, which involves finding the optimal prompts to solve each downstream task.

Recent studies have explored the application of prompt learning in depression detection, particularly in low-resource scenarios. For example, Guo et al. [16] proposed a prompt-based topic modeling method for depression detection, leveraging personalized topic prompts to capture clinically relevant patterns from limited annotated data. Their approach demonstrated significant improvements in model generalization by dynamically adapting prompts based on individual characteristics. Similarly, Gong and Poellabauer [17] introduced a topic modeling-based multimodal depression detection framework, where prompts were used to align textual and audio features, enhancing the model's ability to capture subtle emotional cues. These studies highlight the potential of prompt learning in addressing data scarcity and improving the interpretability of depression detection models.

Despite its promise, prompt learning in depression detection faces several challenges. First, the selection of optimal prompts is highly sensitive to the specific context and individual characteristics, requiring careful design and adaptation [24]. Second, existing methods often struggle to balance the trade-off between personalization and generalization, as overly specific prompts may hinder knowledge sharing across individuals [25]. Third, the integration of prompts with multimodal data (e.g., text, audio, and video cues) remains an open challenge.

2.3. Application of wide & deep model in depression prediction

The Wide & Deep model, initially proposed by Google, has demonstrated its effectiveness across various domains, including recommendation systems [18], natural language processing [26], and advertising [27]. As shown in Fig. 1, the model's architecture combines the memorization capabilities of the Wide component with the generalization abilities of the Deep component, offering a robust framework for tackling personalized challenges. In recommendation systems, the Wide & Deep model plays a key role in capturing users' historical preferences through the Wide component, while the Deep component learns new patterns from data. This dual approach significantly enhances the accuracy and generalizability of recommendation algorithms [26]. For example, in e-commerce, the model can remember users' past purchases and recommend new products based on emerging trends [28].

However, applying the Wide & Deep model to depression prediction requires a deep understanding of clinical and psychological factors, as well as the ability to handle sparse, high-dimensional data derived from personal information such as age, gender, and educational background. The Wide component memorizes these individual characteristics, which are essential for generalized depression prediction. To tackle the depression prediction task, the Deep component must process multimodal data, including text, audio, and video cues. The deep neural network needs to identify patterns indicative of depressive symptoms from interview records, audio files, and facial expressions in videos. By integrating individual data with multimodal features, the Wide & Deep model can effectively predict depression severity. Its ability to combine historical data with complex multimodal signals makes it a promising tool for generalized depression prediction.

3. Theory

TPMN combines personal information with video images and audio details based on textual topic prompts to more effectively predict depression severity. TPMN consists of a Wide component and a Deep component, as shown in Fig. 2. Personal information is remembered through the Wide module, and the prompt pool module is constructed by a topic extraction model that extracts interview topics from the text records, forming key–value pairs for the prompt pool. The topic prompt query module dynamically selects appropriate topic prompts for different text inputs, providing corresponding topic prompts for subsequent Deep Text prompting. The Deep Text prompting module provides relevant prompts for the text model. The prompt coupling function module transforms the prompts from the topic prompt query module into prompts for deep visual prompting and deep auditory prompting, achieving dynamic interaction between modalities. The Deep Visual prompting module and Deep Auditory prompting module provide relevant prompts for the visual and audio models, respectively. The joint training module combines the Wide part and the Deep part to achieve generalized multimodal depression recognition for different personal information. The Wide component utilizes generalized linear models (GLMs), which are well-suited for modeling simple, sparse features such as personal information (e.g., age, gender, education level). GLMs excel at capturing cross-feature interactions (e.g., "sex = woman and age = 20–30"), which are crucial for personalized depression prediction. In contrast, learning such simple features using the Deep component may lead to overfitting, as deep neural networks tend to over-parameterize sparse data, which reduces their generalization ability.

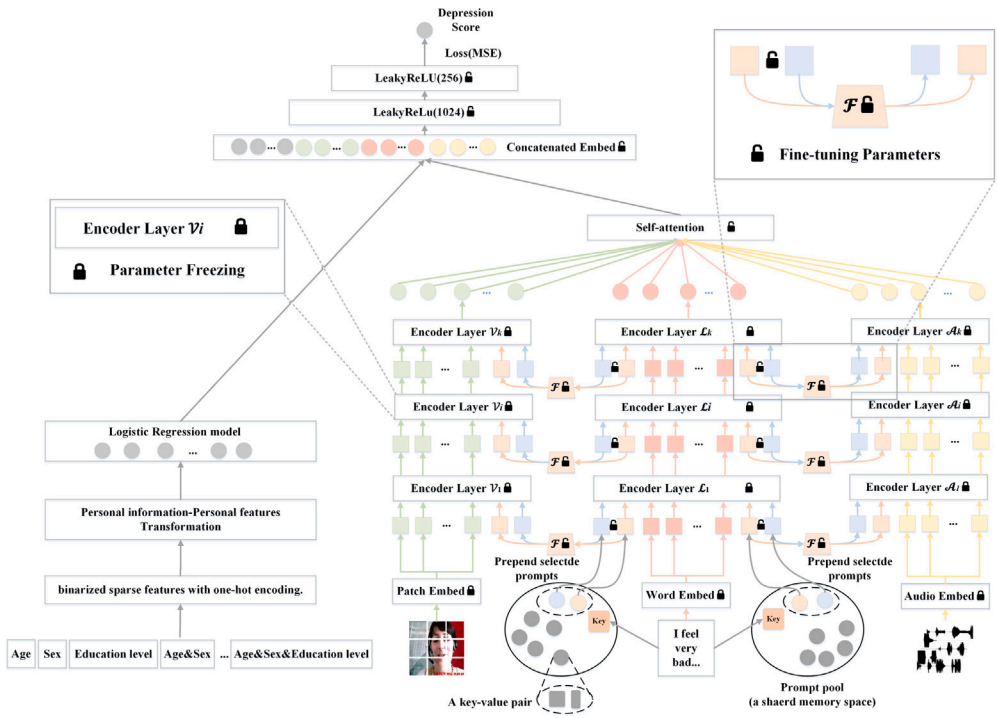

Fig. 2. Topic prompts multimodal net.

While the Wide component focuses on learning personal information, the Deep component is dedicated to extracting and integrating complex multimodal features, ensuring a balanced approach that mitigates overfitting while enhancing generalization. The Deep component, built on deep neural networks, excels at extracting complex, hierarchical features from multimodal data, such as text, audio, and video cues. This capability is essential for tasks such as topic prompt generation, cross-modal feature alignment, and dynamic interaction between modalities. The Deep component's architecture, particularly its Transformer layers, is designed to capture long-range dependencies and contextual information across modalities. For example, language-specific deep prompts are dynamically adjusted based on the input text, while deep visual prompts are optimized for facial expression analysis, enabling the model to adapt to individual differences and cultural variations. The Deep component's ability to learn generalized representations allows it to effectively integrate knowledge from the prompt pool, facilitating knowledge sharing among similar individuals while maintaining independence for dissimilar cases. This design ensures that the model can generalize well to unseen data, even in low-resource settings. The Deep component includes a prompt pool, topic prompt queries, deep language prompts, deep visual prompts, deep audio prompts, and prompt coupling functions to achieve generalization capabilities. Finally, a joint training loss is employed to train the model.

3.1. Personal information memory

The input information includes personal details such as gender, age, and education level, as shown in Fig. 3. Binary features are extracted from the input personal information. For instance, the binary feature "sex = woman" is assigned a value of 1 to indicate that the subject is female. By performing cross-feature transformation on sparse features, such as ("sex = woman and age = 20–30"), TPMN can effectively learn the information, with a value of 1 indicating that the subject is a female aged between 20 and 30. Generalized linear models are typically trained on sparse features with one-hot encoding, in the form of $y = \mathbf{w}^{\top}\mathbf{x} + b$. Where, y represents the predicted outcome, $x = [x_1, x_2, \ldots, x_d]$ is the vector of d features obtained by embedding personal information, $w = [w_1, w_2, \ldots, w_d]$ are the model parameters, and b is the bias term. The set of features includes both the original input features and the features after cross-transformation. Specifically, the cross-transformed features are generated by combining multiple original features to capture their interactions, which are crucial for personalized depression prediction. These features are formally defined as follows:

$$\varnothing_k(x) = \prod_{i=1}^{d} x_i^{c_{ki}}, \quad c_{ki} \in \{0, 1\} \tag{1}$$

where c_{ki} is a Boolean variable that Equals 1 if the ith feature is included in the kth transformation ϕ_k, and 0 otherwise. For binary-crossed features, it equals 1 when all constituent features (e.g., "sex = woman and age = 20–30") are 1, and 0 otherwise. This captures the interaction between binary features, which can influence predictive patterns and outcomes. For example, in males aged 10–20, topics related to family and education become key to identifying depression within this group. This interaction between age and gender can reveal specific risk factors or behavioral patterns linked to depression in young males. Similarly, ternary groups that combine age, gender, and education level have been incorporated, introducing non-linearity into the generalized linear model. These groups enhance TPMN's capacity to capture complex relationships within the data, thereby improving its predictive accuracy.

3.2. Prompt pool

3.2.1. Construction of the prompt pool

The prompt pool is designed to enable knowledge sharing among individuals with similar attributes while preserving knowledge independence across instances. To achieve this, the Wide component stores personal information, while the prompt pool contains structured knowledge. This knowledge can be flexibly organized as inputs for the Deep component, supporting various forms of personal information. The definition of the prompt pool is as follows:

$$P = \{P_1, P_2, \ldots, P_M\}, \quad M = \text{total\#of prompts} \tag{2}$$

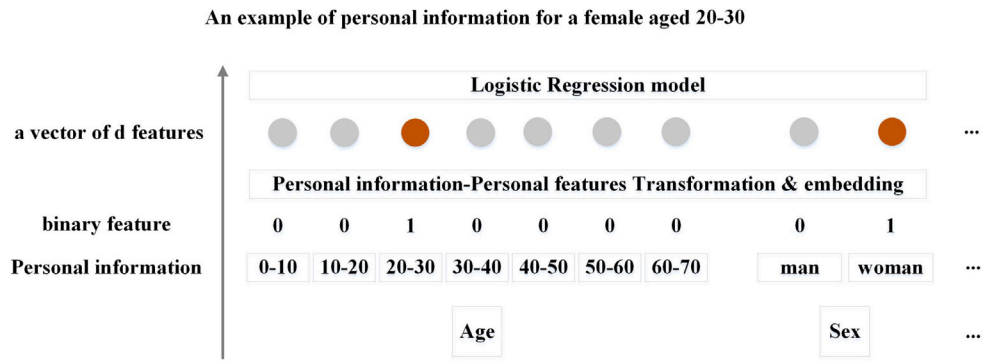

Fig. 3. Schematic diagram of the Wide part structure.

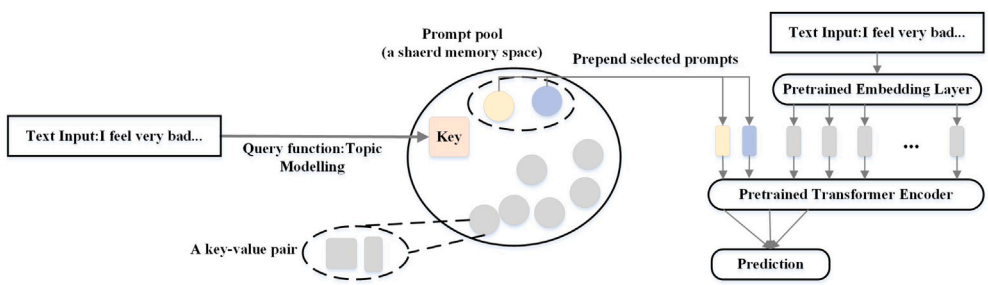

Fig. 4. Schematic diagram of topic pool construction and querying.

where, $P_j \in \mathbb{R}^{L_p \times D}$ represents a single topic prompt with a token length of L_p. The input x is the interview text, and $x_e = f_e(x)$ denotes the embedded features of the text input after passing through the deep language module, which has the same embedding size D as the topic prompt. In the rest of this paper, prompt stands for topic prompt. Let $\{s_i\}_{i=1}^{N}$ be a subset of N topic indices from the set [1, M], and then adjust the input embeddings as follows:

$$x_p = \left[P_{s_1}; \ldots; P_{s_N}; x_e\right], \quad 1 \le N \le M \tag{3}$$

where ';' denotes concatenation along the token length dimension. The prompts can be freely combined, allowing them to encode knowledge together (for example, an interview may encompass two or more topics) for model processing. The objective is to implement a more detailed knowledge-sharing mechanism on the instance level through prompt integration, where prompt information is aligned with topics relevant to individuals. Consequently, similar individual inputs share more common topic prompts, and vice versa. As shown in Fig. 4, the prompt pool provides the deep language module with the relevant prompt information.

3.2.2. Topic-prompted queries

A key–value pair-based query strategy was developed to dynamically select appropriate topic prompts for varying inputs, as illustrated in Fig. 5. This strategy helps match input data to the most relevant prompts, enhancing TPMN's ability to tailor its responses to the specific characteristics of each input.

Each prompt is associated with a learnable key, denoted as(k_1, P_1), $(k_2, P_2), \ldots, (k_M, P_M)$, where $k_i \in \mathbb{R}^{D_k}$. The set K = $\{k_i\}_{i=1}^{M}$ is derived from applying Latent Dirichlet Allocation (LDA) topic modeling to the entire corpus of interview texts. The selection of LDA over other topic modeling methods, such as multi-grain topic models [29], dynamic topic models [30], and BERTopic [31], is based on its probabilistic generative mechanism, which is well-suited for inferring latent topics from structured and limited data. Clinical interviews, such as those in the AVEC2013/2014 dataset, typically involve a limited number of topics, and subjects do not proactively initiate topics. During clinical interviews, topics are often initiated using a small set of fixed sentences (1–3 sentences per topic). LDA's probabilistic generative mechanism allows it to model the distribution of topics within documents and the distribution of words within topics, making it particularly effective for extracting clinically relevant patterns from structured interview texts. As demonstrated in [18], LDA has been widely validated for topic modeling in Multimodal Depression Detection. Its effectiveness in capturing latent topics from structured text data further supports its suitability for this task. Given these advantages, LDA was chosen as the most appropriate method for topic modeling in this study, enabling the effective extraction of interview topics and the dynamic selection of topic prompts through a query–key matching process. The design philosophy is to enable input instances to autonomously determine the selection of topic prompts through a query–key matching process. To this end, we introduce a query function q: $\mathbb{R}^D \rightarrow \mathbb{R}^{D_k}$, which encodes the input interview text x into the same dimension as the keys. Additionally, q should be a deterministic function without learnable parameters for different tasks.

The LDA model is utilized directly to obtain the query features for the current interview text, defined as q(x) = LDA(x). This model is capable of extracting relevant topic features from the input text x, projecting texts from various tasks into a topic feature space. LDA model is conducive to facilitating model convergence.

Let $\gamma : \mathbb{R}^{D_k} \times \mathbb{R}^{D_k} \rightarrow \mathbb{R}$ represent a function for evaluating the match between a query and the prompt keys, where the cosine distance has been identified as an effective measure for this purpose. Given an input x, we use q(x) to simply solve the following objective to find the top N keys:

$$K_x = \underset{\{s_i\}_{i=1}^{N} \subseteq [1,M]}{\operatorname{argmin}} \sum_{i=1}^{N} \gamma(q(x), k_{s_i}) \tag{4}$$

where, K_x denotes a subset of the top N keys selected specifically for x from K. The value of N is determined by the number of interview sentences, with one topic selected for every three sentences. Specifically, if the number of sentences is S, then N is calculated as N = $\lceil S/3 \rceil$, where $\lceil \cdot \rceil$ denotes the ceiling function, which rounds up to the nearest integer.

For example, when the number of sentences is 5, N is set to 2. This key–value strategy design decouples the learning of the query mechanism from the prompt learning process. Moreover, the query prompts are conducted on a per-interview instance basis, meaning that this method does not require explicit interview topics during training, nor does it require interview topic labels during testing.

3.2.3. Prompt pool optimization

In each training step, after selecting N prompts according to the aforementioned query strategy, these prompts are fed as inputs. The resulting embedded features, denoted as x_p, are then fed into the remaining part of the pre-trained model, f_r, which consists of the attention mechanism layers in the transformer and the final classifier $g_\emptyset$ parameterized by $\emptyset$, where $g_\emptyset$ is a fully connected layer. Overall, we aim to minimize the end-to-end training loss function:

$$\min_{P,K,\phi} \mathcal{L}\left(g_\phi\left(f_r^{\text{avg}}(x_p)\right), y\right) + \lambda \sum_{K_x} \gamma(q(x), k_{s_i}), \text{s.t.} \tag{5}$$

where, $f_r^{\text{avg}} = \text{AvgPool}(f_r(x_p)[0 : NL_p, :]), i.e.$,where AvgPool represents average pooling, and $(f_r(x_p)[0 : NL_p, :])$ represents training only on the N prompt features. In Eq. (4), the output hidden vectors corresponding to the prompt NL_p positions are averaged. The first term is the loss of the joint training module, which will be presented in the subsequent methodology. Note that the joint training loss and the prompt loss are optimized separately. The second term is a prompt loss that pulls the selected prompts closer to their corresponding query features. This separation allows the model to balance the overall training objective while maintaining the specificity of the prompts.λ is a scalar used to weight the losses. This module can train the feature expressions of the N selected prompts and ultimately output the optimal prompt feature expressions.

3.3. Multimodal deep prompting

3.3.1. Deep text prompting

The text encoder generates a feature representation of the text description by tokenizing words and projecting them into word embeddings $W_0 = \left[w_0^1, w_0^2, \dots, w_0^N\right]$. In each step, W_i is input into the $(i+1)$th Transformer layer L_{i+1} of the text encoding branch, $\left[W_i\right] = L_i(W_{i-1})$. To learn prompts related to depression discrimination in interview texts, language model introduces N learnable topic prompts $\{P^i \in \mathbb{R}^{L_p \times D}\}$ from the prompt pool, where i ranges from 1 to N. The form of the input embeddings is $[P_0, W_0]$, where $W_0 = [w^1, w^2, \dots, w^N]$ corresponds to the fixed input tokens, and $P_0 = [P_{s_1}; \dots; P_{s_N}]$ corresponds to the optimal prompt features queried from the prompt pool. In each transformer block of the language encoder L_i, new learnable tokens are further introduced up to a specific depth J.

$$[_, W_i] = L_i([P_{i-1}, W_{i-1}]), \quad i = 1, 2, \dots, J \tag{6}$$

where $[\cdot,]$ denotes the concatenation operation. Shallow networks struggle to capture complex prompt information, while deep networks, due to their stronger abstraction capabilities and more complex nonlinear transformations, can better process and convey these prompt messages. Therefore, in the L_1 to L_j layers of the deep text prompting module, the optimal textual topic prompts for each layer input are not propagated; instead, the approach is to re-input the optimal textual topic prompts for each layer. In the L_j to L_k layers of the deep text prompting module, the optimal textual topic prompts for each layer's input are transmitted between layers, meaning that the optimal textual topic prompts from layer L_j, after being encoded, are then input into layer L_{j+1}.

$$[P_j, W_j] = L_j([P_{j-1}, W_{j-1}]), \quad j = J+1, \dots, K \tag{7}$$

$$z = \text{TextProj}([P_K, W_K]) \tag{8}$$

As shown in the middle section of Fig. 5, the final output text representation z is obtained by projecting the text embedding corresponding to the last token of the last Transformer block L_K into a common multimodal feature space through TextProj. The projection components, such as TextProj, ImageProj, and AudioProj, are responsible for projecting the class token from the last Transformer layer into a common multimodal feature space.

3.3.2. Deep visual prompting

Image Encoding: The image encoder V consists of K Transformer layers $\{V_i\}_{i=1}^K$, which divides the input image I into M fixed-size patches. These patches are projected into Patch Embedding $E_0 \in \mathbb{R}^{M \times d_v}$. The patch embeddings E_i, along with the learnable class (CLS) token c_i, are input into the $(i+1)$th Transformer block V_{i+1}, and are processed sequentially through K Transformer blocks. CLS markers can help TPMN distinguish between depressive and non-depressive traits with the same personal information. $[c_i, E_i] = V_i[c_{i-1}, E_{i-1}]$ for $i = 1, 2, \dots, K$. To obtain the final image feature representation x, the class token c_k from the last Transformer layer (V_K) is projected into a common multimodal feature space through ImageProj.

$$x = \text{ImageProj}(c_K)x \in \mathbb{R}^{d_{v1}} \tag{9}$$

Similar to deep Text prompting, the utilized model is a Vision Transformer pre-trained model. Within the visual model, N learnable topic prompts $\{\tilde{P}_1^i \in \mathbb{R}^{d_v}\}_{i=1}^N$ are introduced, which are coupled with the input image tokens and generated by the prompt coupling module. Throughout the deeper Transformer layers of the image encoder (V), new learnable tokens are introduced up to depth J. The feature from the last Transformer layer is projected into a common feature space through ImageProj, as illustrated in the left part of Fig. 5.

$$[c_i, E_i, \cdot] = V_i([c_{i-1}, E_{i-1}, \tilde{P}_1^{i-1}]), \quad i = 1, 2, \dots, J \tag{10}$$

$$[c_j, E_j, \tilde{P}_1^j] = V_j([c_{j-1}, E_{j-1}, \tilde{P}_1^{j-1}]), \quad j = J+1, \dots, K \tag{11}$$

$$x = \text{ImageProj}([c_k, E_k, \tilde{P}_1^k]) \tag{12}$$

3.3.3. Deep audio prompting

Akin to Deep Text prompting, TPMN employed is the Wav2vec2.0 pre-trained model. Within the audio model, N learnable topic prompts $\{\tilde{P}_2^i \in \mathbb{R}^{d_a}\}_{i=1}^N$ by the prompt coupling module, which are combined with the input audio token. Additionally, new learnable tokens are introduced in the deeper Transformer layers of the audio encoder (A) up to depth J, as depicted in the right part of Fig. 5. The pre-trained model utilized is the Wav2vec2.0 model. Within the audio model, N learnable topic prompts $\{\tilde{P}_2^i \in \mathbb{R}^{d_a}\}_{i=1}^N$ are introduced, which are coupled with the input audio tokens and generated by the prompt coupling module.

$$[a_i, A_i, _] = S_i([a_{i-1}, A_{i-1}, \tilde{P}_2^{i-1}]), \quad i = 1, 2, \dots, J \tag{13}$$

$$[a_j, A_j, \tilde{P}_2^j] = S_j([a_{j-1}, A_{j-1}, \tilde{P}_2^{j-1}]), \quad j = J+1, \dots, K \tag{14}$$

$$a = \text{AudioProj}([a_k, A_k, \tilde{P}_2^k]) \tag{15}$$

where, a learnable class token is employed, functioning similarly to the CLS token in visual prompting. Consistent with visual prompting, additional learnable tokens are introduced in the deeper Transformer layers of the audio encoder (A), up to depth J, as indicated in the right part of Fig. 5.

3.3.4. Prompt coupling function

Adjusting prompts is essential for the multimodal approach, requiring the tuning of the video, audio, and text branches. A common method involves combining deep video prompt P_1, audio prompt $\tilde{P}_2$,

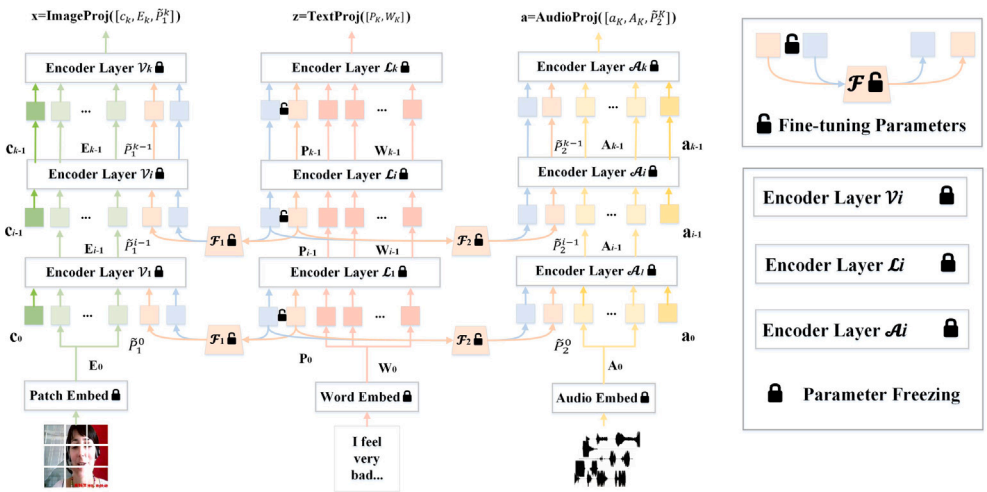

Fig. 5. Schematic diagram of the structure of the Deep section.

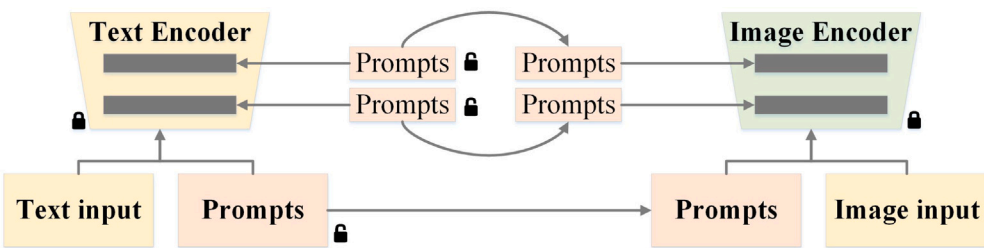

Fig. 6. Dynamic inter-modal interaction between text and image.

and text prompt P, which are learned concurrently during training. While this ensures prompt completeness, it lacks synergy between the video, audio, and text branches, and fails to promote interaction between the text and video/audio branches in learning contextually relevant prompts, a crucial aspect for enhancing TPMN's ability to integrate multimodal information effectively. A branch-aware multimodal prompting method is proposed to integrate prompts effectively across modalities and to jointly adjust TPMN's video, audio, and text branches. Text prompt tokens are incorporated into the text branch up to the Jth Transformer block. To foster mutual synergy between Text–Video (T–V) prompts, the video prompt P_1 is derived through a projection from the text to video prompt space, referred to as the T–V coupling function $F_1(\cdot) = \mathrm{ReLU}(\cdot)$, such that $P_1^k = F_k(P_k)$ maps the text prompt into the video prompt space. The coupling function is instantiated as a 768×768 linear layer, followed by a ReLU function to obtain the coupled prompt information. V_i denotes the ith visual nonlinear layer with cross attention and V_j denotes the jth visual nonlinear layer with cross attention. In the $T_1 - T_j$ layers, the mapped optimal text topic prompts are re-input to each layer in a way, as shown in Eq. (16). In the $T_j - T_k$ layers of the deep text module, the mapped optimal text topic prompts are inputted to each layer for layer-to-layer transfer. Specifically, the mapped optimal text topic prompts in the T_j layer are encoded and re-inputted to the layers, as shown in Eq. (17). The final output $x = \mathrm{ImageProj}(c_K)$ is shown in Eq. (18). This mechanism acts as a conduit between the two modalities, facilitating the exchange of gradients and enhancing the integrated learning of prompts across text and video branches (see Fig. 6).

$$[c_i, E_i, _] = V_i([c_{i-1}, E_{i-1}, F_1^{i-1}(P_{i-1})]), \quad i = 1, 2, \ldots, J \tag{16}$$

$$[c_j, E_j, P_1^j] = V_j([c_{j-1}, E_{j-1}, P_1^{j-1}]), \quad j = J+1, \ldots, K \tag{17}$$

$$x = \mathrm{ImageProj}(c_K) \tag{18}$$

Unlike independent V–T prompts, explicitly conditioning P_1 on P helps to learn prompts in the shared embedding space between the two branches, thereby enhancing T–V mutual synergy.

To achieve mutual synergy between Text–Audio (T–A) prompts, the audio prompt P_2 is derived through a projection from the text to audio domain, a process $\tilde{P}_k^2 = F_k^2(P_k)$ denoted as the T–A coupling function $F_2(\cdot) = \mathrm{ReLU}(\cdot)$. The coupling function is instantiated as a 768×768 linear layer, followed by a ReLU function to obtain the coupled prompt information. Similarly, S_i denotes the ith audio nonlinear layer with cross-attention, and S_j denotes the jth audio nonlinear layer with cross-attention. In this paper, a similar mechanism to the prompting is used for audio prompting. Eqs. (19)–(21) represent the prompt feature transfer under the dynamic coupling mechanism, with the final output $x = \mathrm{AudioProj}(c_K)$. This mechanism transforms the text prompt into the audio prompt space, resulting in a mapping from the text modality to the audio modality. The coupling function operates as a linear layer that projects the input from d_l-dimensional to d_k-dimensional, thereby facilitating the integration of audio prompts with language prompts. This linear transformation enables TPMN to align the audio features with the language features, enhancing the multimodal learning process.

$$[a_i, A_i, _] = S_i([a_{i-1}, A_{i-1}, F_2^{i-1}(P_{i-1})]), \quad i = 1, 2, \ldots, J \tag{19}$$

$$[a_j, A_j, \tilde{P}_2^j] = S_j([a_{j-1}, A_{j-1}, \tilde{P}_2^{j-1}]), \quad j = J+1, \ldots, K \tag{20}$$

$$x = \mathrm{AudioProj}(c_K) \tag{21}$$

Unlike independent A–T prompts, explicitly conditioning $\tilde{P}_2$ on P facilitates the learning of prompts within the shared embedding space between the two branches, thereby enhancing T–A mutual synergy.

3.3.5. Jointly optimized

The Wide and Deep components serve as inputs, as shown on the left and right sides of Fig. 2. Through fully connected layers and

LeakyReLU activation functions, the output is used as the prediction result, which is then fed into a common mean squared loss function for joint training, as shown in Eq. (23). It is important to distinguish between joint training and ensemble methods. In ensemble methods, individual models are trained separately, unaware of each other's models or predictions; their predictions are only combined during inference, not during training. In contrast, joint training optimizes all parameters by considering both the wide and deep components and their weighted sum, enabling TPMN to select the most suitable topic prompts while taking personal information into account.

Joint training of the Wide & Deep model is performed through mini-batch stochastic optimization, allowing the simultaneous backpropagation of gradients from the output to both components of TPMN—the wide part and the deep part. During the experiments, the Adam algorithm was chosen as the optimizer for both parts of TPMN. For the task of depression score prediction, TPMN's predictive output is formulated as follows:

$$P(Y|x) = \sigma\left(L_f\left(w_{\text{wide}}^T[x, \phi(x)] + w_{\text{deep}}^T a^{(l_f)}\right) + b\right) \tag{22}$$

$$\text{Loss} = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - P(Y|x_i)\right)^2 \tag{23}$$

where, Y is the predicted score label, $(\cdot)$ represents the LeakyReLU activation function, $\phi(x)$ denotes the cross-features of the raw feature x, and b is the bias term. w_{wide} is the vector of weights for the wide model, and w_{deep} is the weight applied to the final activation function $a^{(l_f)}$. L_f represents the fully connected layer, where $L_f = w_l^T Z + b_l$, with $Z = w_{\text{wide}}^T[x, \phi(x)] + w_{\text{deep}}^T a^{(l_f)}$ indicating the input.

4. Experimental methods

4.1. Data base

In this study, we conducted experiments using two publicly available datasets and one self-collected dataset. The public datasets are the AVEC2013 and AVEC2014 depression datasets.

The AVEC2013 depression dataset is a subset of the Audio/Visual Depression Corpus (AViD corpus), which records each subject completing 14 tasks using a webcam and microphone, as instructed on the computer screen. Subjects' ages ranged from 18 to 63 years, with a mean age of 31.5 years and a standard deviation of 12.3 years. All subjects are German speakers. The dataset consists of 150 video clips, divided into training, validation, and test sets, each containing 50 samples. The video recordings were made at a resolution of 640 × 480 pixels, with a frame rate of 30 frames per second.

The AVEC2014 dataset is a subset of AVEC2013, sharing similarities in collection environment, age distribution, and text characteristics. It includes two tasks: "North Wind" and "Free Form". In the "North Wind" task, participants are required to read an excerpt from the German fable "Die Sonne und der Wind". In the "Free Form" task, participants respond to questions in German, such as "What is your favorite dish?" or "Describe a sad memory from your childhood". The dataset consists of 150 audio recordings from 84 participants, evenly divided into training, validation, and test sets, with durations ranging from 6 s to 4 min.

In this study, we collected a scientifically rigorous and standardized depression dataset, currently comprising 43 individuals, including 23 patients diagnosed with depression and 20 control subjects. The patients with depression range in age from 18 to 60, with most between 20 and 40 years. The diagnosis of depression must be confirmed by at least two attending psychiatrists or at least on. All participants signed an informed consent form, and the project passed ethical review.

The dataset encompasses various research topics, including learning, work, marriage, and family, capturing participants' interests, emotional states, stress levels, and video/audio recordings. For example, a series of structured interview questions on marriage have been developed, such as: "What do you think of your spouse?" and "Are you satisfied with your marital life?" A response indicating sadness due to the financial and emotional burden on their spouse suggests guilt.

Table 1
Interview topics and their structured question design.

Topic	Structured questions
Work	What do you think of your job and how does it affect your life?
Matrimonial	How long have you been married to your partner? What do you think your partner is like?
Socialize	How often do you connect with friends or family? How do you usually feel when you interact with others?
Hobbies	What leisure activities do you enjoy in your spare time? How do you think your hobbies affect your life?
Sleep condition	How has your sleep been lately? Are you experiencing any sleep problems?
Emotional state	How have you been feeling lately? Is there anything specific that would make you happy?
Dietary situation	Are your meal times regular? What do you think of your diet?

To ensure a comprehensive assessment of patients' depressive symptoms, severity, potential causes, and impacts, structured questions were developed to gather all relevant information for comparison and analysis. These questions are framed in clear, professional, and empathetic language, adhering to sensitivity guidelines for mental health communication. The structured interview format maintains consistent question wording and order.

Informed by clinical indicators and depressive symptoms — such as persistent sadness, despair, cognitive dysfunction (e.g., reduced attention and decision-making difficulties), and appetite/weight changes — structured questions were developed around topics like work, marriage, and social interaction, as shown in Table 1.

For our self-collected dataset, the chi-squared test was used to evaluate the association between categorical variables and group status (patients vs. control group), as shown in Table 2. Additionally, independent samples t-tests were conducted to compare the mean values of continuous variables between the patient and control groups, with results presented in Table 3.

Significant differences were observed in height and years of education between the groups $p < 0.01$. Educational disparity: The control group, consisting entirely of university students, exhibited significantly higher education levels (16.4 vs. 12.1 years, $p < 0.00001$), which may introduce selection bias. Age difference: No significant associations were found between age or any other categorical variables and group membership $p > 0.05$. Matched demographics: The distributions of gender, smoking, and alcohol use were nearly identical between the groups. Educational bias: The homogeneity of the control group, consisting solely of university students, limits the generalizability of the findings. Future studies will recruit controls with more diverse educational backgrounds (ongoing data collection). Despite demographic differences, no baseline imbalances were observed in core clinical variables.

4.2. Data preprocessing

Implementing a Generalized multimodal depression severity prediction method involves comprehensive preprocessing of the interview records from depression patients, to extract features associated with depressive topics across various modalities, as shown in Fig. 7.

4.2.1. Text modality preprocessing

Speech-to-Text and Topic Segmentation: The initial step involved using advanced iFlytek speech recognition technology to transcribe verbal exchanges from the interview videos into clear text. Subsequently, the texts were segmented according to the specific topics discussed

Table 2
Independent samples t-tests were performed on the means of continuous variables in the patient and normal control groups.

Variable	Patients	Normal	$\mathcal{X}^2$	p-value	Significance
BMI	23.08 ± 3.79	21.48 ± 3.15	1.491	0.1437	n.s
Education level	4.96 ± 1.72	7.8 ± 0.62	−7.01	0	significance
Smoking	0.22 ± 0.42	0.05 ± 0.22	1.59	0.1196	n.s
Years of education	12.13 ± 3.12	16.15 ± 0.67	−5.635	0	significance
Income	3.17 ± 1.67	2.55 ± 2.01	1.111	0.2729	n.s
Marital status	1.57 ± 0.9	1.05 ± 0.22	2.502	0.0164	n.s
Number of children	0.74 ± 1.1	0.05 ± 0.22	2.758	0.0086	n.s

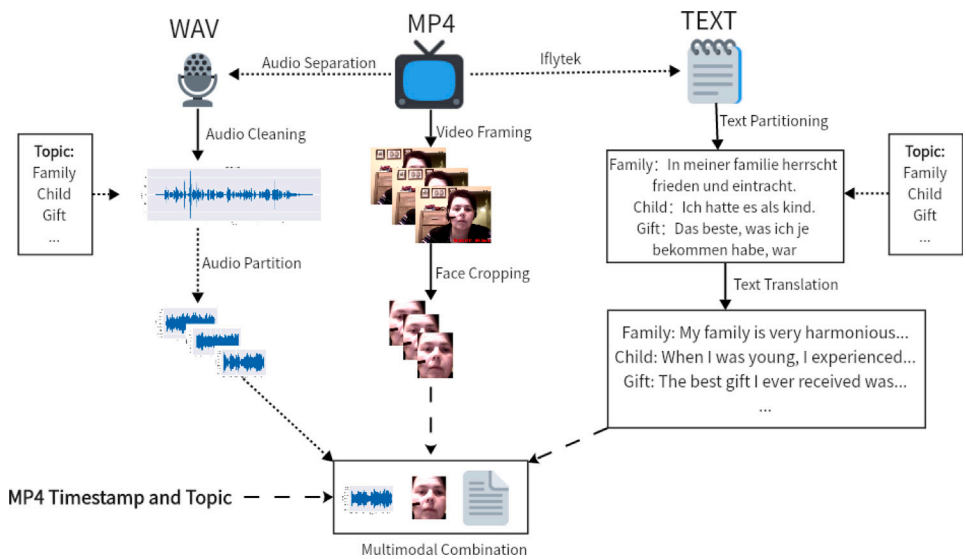

Fig. 7. Data preprocessing methods.

Table 3
A chi-square test was used to assess the association between categorical variables and group (patients vs. normal controls).

Variable	Patients	Normal	t_statistic	p_value	Significance
Waist	85.96 ± 10.79	78.35 ± 10.77	2.308	0.0261	n.s.
Height	162.43 ± 7.03	172.4 ± 8.86	−4.108	0.0002	significance
Gender	1.7 ± 0.47	1.4 ± 0.5	1.991	0.0532	n.s.
Employment	2.0 ± 1.13	1.15 ± 0.37	3.221	0.0025	n.s.
Drinking	0.22 ± 0.42	0.55 ± 0.51	−2.34	0.0242	n.s.
Weight	60.74 ± 9.28	64.5 ± 13.97	−1.052	0.2989	n.s.
Age	30.05 ± 11.80	23.72 ± 1.56	2.25	0.0303	n.s.

in the interviews, enabling efficient feature extraction and pattern analysis. Since most pre-trained language models are primarily based on English corpora, the Chinese and German texts were translated into English. This adaptation was crucial for enhancing the model's applicability and improving predictive

4.2.2. Video modality preprocessing

Video Framing and Facial Focus: Efficient video framing techniques were employed to sample the interview videos at a rate of 30 frames per second, producing a sequence of image frames. To capture subtle variations in facial expressions, the facial regions in each frame were cropped, removing background distractions and focusing on the primary emotional expressions.

4.2.3. Audio modality preprocessing

Audio Segmentation and Standardization: The audio signals were segmented based on the thematic content of the interviews, with particular focus on dialogue segments related to depressive symptoms. Audio segments shorter than one second were padded with silence to standardize their length, ensuring that all audio samples were consistently adjusted to a duration between 2 and 30 s. This process preserved data integrity and consistency.

Achieving multimodal data alignment involved implementing a timestamp-based marking method. During the preprocessing phase, each data block — whether it be text paragraphs, image frames, or audio segments — was assigned a precise timestamp. This methodology enabled the synchronization of corresponding text, image, and audio data during the data fusion process, based on the assigned timestamps. Timestamp-based alignment ensures that text, audio, and video data corresponding to the same emotional state are processed together. For instance, a facial expression captured in a video frame is aligned with the corresponding speech segment and its transcribed text. This synchronization allows the model to learn more accurate and meaningful relationships across modalities, thereby improving its ability to detect depression-related cues. By aligning multimodal data, the model gains a better understanding of the context surrounding emotional expressions. For example, a sad facial expression, accompanied by a slow, monotone voice and negative textual content, provides a coherent signal for depression detection. Without alignment, the model might misinterpret isolated features, resulting in reduced performance. During training, timestamp-based alignment ensures that the model learns from coherent multimodal inputs, reducing noise and enhancing convergence. During inference, it enables the model to generate more accurate predictions by leveraging synchronized features. Our experimental results demonstrate that this alignment strategy contributes to TPMN's superior performance on both the AVEC2013 and AVEC2014 datasets, yielding an MAE of 3.01 and 2.70, respectively.

4.3. Construction of the prompt pool

Most topic modeling methods employ probabilistic generative models to define the document generation mechanism and infer latent topics. One of the most widely used techniques is Latent Dirichlet

Table 4
List of extracted topics from the AVEC2013/AVEC2014 and our datasets.

Ind.	Topic	Ind.	Topic
1	An indescribable picture	13	Grandma's cooking
2	Beautiful summertime	14	Growing up under family
3	Best experience	15	Guilty affects behavior
4	Best gift	16	Mom's cooking
5	Cannot remember any gifts	17	No idea
6	Cannot remember best gift	18	Nothing to say
7	Childhood favorite food	19	Pleasant trip to America
8	Childhood favorite game	20	Prefer handcrafted to games
9	Children's bicycles	21	Recalling grandma's cooking
10	Christmas atmosphere	22	Road to destination
11	Diabetes for childbirth	23	Story about wind and sun
12	Elderly woman's actions	24	Struggle against alcoholism

Allocation (LDA) [32]. There are several reasons for using LDA in topic modeling of clinical interview transcripts. First, the number of topics in clinical interviews is limited, and participants do not actively initiate topics, which makes more general methods, such as multi-grain and dynamic topic models, less effective in capturing their characteristics. Second, the way topics are introduced in interviews is relatively fixed and predictable.

To perform topic modeling, a simple approach was applied to the interview texts, starting with the creation of a preliminary sentence dictionary. This involved iterating through all the interview transcripts and cataloging all non-redundant sentences. The preliminary dictionary was then manually refined, removing sentences that did not introduce new topics, such as those that merely affirmed or repeated information without initiating a topic (e.g., "That's good".). Following this, the LDA (Latent Dirichlet Allocation) method was applied to extract topics from the cleaned interview texts. The resulting topics are presented in Table 4, and all topics listed therein form the prompt pool.

4.4. Training setting

The experiments were conducted in a distributed training environment using four NVIDIA V100 GPUs (32 GB VRAM each), interconnected via NVLink. For the AVEC2013 dataset, we employed a global batch size of 128 (32 per GPU), completing training in 6.2 h with peak memory utilization of 28.4 GB per GPU. The 4-GPU configuration achieved a linear speedup of 3.57× through gradient accumulation and mixed-precision optimization, reducing the average per-epoch training time from 8.2 min (single GPU) to 2.3 min.

Ablation studies on the AVEC2014 dataset used an increased per-GPU batch size of 128 (global batch size of 512), maintaining stable memory usage at 24.8 GB and completing training in 2.1 h, with throughput reaching 1280 samples/second. Distributed communication analysis showed that AllReduce operations accounted for 17% of the total training time, mitigated by a 38% data reduction through gradient compression techniques. For inference, single-GPU deployment demonstrated an average latency of 18 ms/sample (batch size = 1), scaling to 2100 samples/second with batch size 128. Cross-dataset zero-shot transfer (AVEC2013→AVEC2014) maintained inference performance with a memory demand of 4.2 GB and consistent processing speed under 25 ms/sample.

Despite using a relatively complex model architecture with the Wide & Deep framework, TPMN has a manageable number of trainable parameters. Specifically, while the total number of parameters is 167,519,374, only 11,285,582 are trainable. The remaining 156,233,792 parameters are frozen during training, which significantly reduces the computational burden during both training and inference. This design choice ensures that TPMN remains computationally efficient and practical for real-world applications, balancing model complexity with high performance and generalization.

The Adam optimizer was used with the following hyperparameters: learning rate $\alpha = 0.001$, first-order momentum decay $\beta_1 = 0.9$, second-order momentum decay $\beta_2 = 0.999$, and numerical stability constant $\epsilon = 10^{-8}$. These settings are widely adopted in the deep learning community for their balance between convergence speed and stability. The adaptive learning rate mechanism of the Adam optimizer allows it to adjust learning rates dynamically based on historical gradient information, making it well-suited for training complex models like the proposed TPMN architecture.

To evaluate the recognition performance of the proposed model, 80% of the data was used as the training set, with experiments conducted on the AVEC2013, AVEC2014 datasets and our self-collected dataset. A validation set consisting of the remaining 20% of the data was held in reserve. In subsequent experiments, 20% of the data was randomly selected as the test set for each trial, with the process repeated five times to obtain an average result, which was considered the final outcome. For the zero-shot experiment, 20% of the data from an untrained dataset was randomly chosen as the test set, following the same procedure of five repetitions to calculate the average result as the final outcome. Furthermore, this approach aligns with the methodology employed by existing state-of-the-art (SOTA) methods in the field. For example, studies such as [33–38] also use random sampling to partition datasets, ensuring that their results are comparable and reproducible. By following this widely accepted practice, we ensure that our experimental setup is consistent with SOTA methods, allowing for a fair comparison of performance. Additionally, to further validate the robustness of our results, we conducted multiple random splits (5-fold cross-validation) and reported the average performance metrics. This approach reduces the risk of overfitting to a specific split and provides a more reliable evaluation of the model's generalization ability.

Ablation experiments were conducted on the AVEC2014 dataset, chosen due to its smaller size, making it more suitable for ablation studies that require multiple model training iterations. The AVEC2014 dataset offers more refined annotations, having been collected after AVEC2013, and could potentially benefit from the experiences and feedback of the earlier dataset, thereby providing higher-quality labeled data. This is especially important for ablation experiments. Although AVEC2014 is a subset of AVEC2013, it includes participants from diverse countries and cultural backgrounds, offering a more varied dataset for ablation experiments.

In the zero-shot experiments, TPMN was trained using the AVEC2013 dataset and subsequently tested for generalization on the AVEC2014 dataset, as well as on an additional dataset not encountered during the training phase. The AVEC2013 dataset was selected for training due to its larger sample size, which allows TPMN to more effectively leverage its personalization capabilities.

5. Results and discussion

5.1. Multimodal performance comparison

Our experiments demonstrate that TPMN achieves superior depression prediction performance when leveraging multimodal interactions, with video and audio modalities consistently outperforming text across all evaluation metrics (MAE: video MAE = 5.69, audio MAE = 6.36, text MAE = 7.06), as shown in Table 6. Depression frequently manifests through non-verbal cues, such as a flattened facial affect (e.g., diminished eye contact or reduced smile dynamics in video) and prosodic irregularities (e.g., a monotonous tone or slowed speech rate in audio). However, these cues are absent in written text, where individuals may consciously or unconsciously conceal their true emotions by carefully controlling their language.

Each modality contributes unique and complementary information to the model. The video modality excels at capturing non-verbal cues, the text modality provides semantic context, and the audio modality adds emotional depth. The coupling functions play a crucial role in

Table 5
Results of comparison with existing methods.

Methods	AVEC2013		AVEC2014	
	RMSE	MAE	RMSE	MAE
2DCNN+pooling method [33] (2020)	5.96	7.97	6.2	7.94
Face2PPG [39] (2021)	7.32	8.73	6.8	8.82
multi-scale-region+3DCNN [34] (2022)	6.19	8.02	6.14	7.98
rPPG and HRV features [35] (2022)	6.43	8.01	6.81	8.63
STA+MAFF [36] (2023)	6.14	8.16	5.12	7.03
NFFT [37] (2023)	–	–	5.81	7.49
audio+video+rPPG [6] (2024)	6.60	9.00	7.05	9.45
MTDAN [38] (2024)	6.14	8.08	6.35	7.93
TPMN	3.01	3.07	2.70	3.65

Table 6
Ablation experiment.

Methods	Datasets	MAE	RMSE
TPMN	AVEC2014	2.7	3.65
NoWide	AVEC2014	10.33	13.60
NoPrompt	AVEC2014	10.46	14.24
NoModal Interaction	AVEC2014	10.12	12.92
Video	AVEC2014	5.69	7.55
Text	AVEC2014	7.06	6.94
Audio	AVEC2014	6.36	7.01

integrating these modalities, ensuring that the model benefits from the full spectrum of available information. This multimodal integration is essential for achieving high performance and generalization in depression detection tasks.

5.2. Depression score prediction

The results demonstrate TPMN's performance compared to SOTA methods, with MAE values of 2.7 and 3.01 on the AVEC2013 and AVEC2014 datasets, respectively, as shown in Table 5. These values indicate that the average discrepancy between TPMN's predicted and actual values is relatively small, reflecting a high level of overall prediction accuracy on the AVEC datasets. The RMSE values of 3.07 and 3.65 further suggest that the average squared prediction errors are low, underscoring the stability and reliability of TPMN's predictions. This suggests that TPMN is capable of capturing features associated with depressive symptoms, which are essential for depression recognition. TPMN's RMSE on AVEC2013 is lower than on AVEC2014, which can be attributed to the larger dataset, enabling TPMN to achieve more stable predictive performance in the presence of extreme values, while also capturing more personalized information in these cases. Meanwhile, TPMN's MAE on AVEC2013 is slightly higher than on AVEC2014, likely due to the increased number of samples and individual variability, resulting in a marginally higher MAE value.

The MAE and RMSE curves for TPMN show an overall decreasing trend during training, with some fluctuations, indicating that the model effectively learns from the multimodal data, as shown in Fig. 8. The validation curves closely follow the training curves, demonstrating that TPMN generalizes well without overfitting. This can be attributed to the balanced design of the Wide & Deep architecture and the dynamic prompt coupling mechanism, which help mitigate overfitting while preserving personalized insights. To further validate TPMN's performance, ablation experiments were conducted.

5.3. Ablation experiment

The results of the ablation experiments are shown in Table 6, where NoWide represents the ablation experiment without the personal information module, NoPrompt represents the ablation experiment without the prompt module, and NoModal Interaction represents the ablation experiment without the coupling function.

5.3.1. Ablation experiment of the wide part

An ablation study was conducted to evaluate the impact of the Wide component in the Wide & Deep model on depression severity prediction performance. Removing the Wide component results in a substantial increase in TPMN's MAE to 10.33, and a corresponding rise in the RMSE to 13.60, indicating a significant degradation in predictive accuracy. Furthermore, in the absence of the Wide component, as shown in Fig. 8, the noWide variant exhibits significantly higher MAE and RMSE values compared to TPMN, with the validation curves diverging from the training curves after just a few epochs. This suggests severe overfitting, as the model fails to utilize personal information for personalized predictions. These results highlight the importance of the Wide component in memorizing individual characteristics and improving generalization. The distribution of predicted and true values during the experimental process is shown in Fig. 9. Analyzing the predicted scores in Fig. 9(b) reveals an interesting phenomenon: when the Wide component is excluded, the predicted scores from TPMN tend to follow a normal distribution centered around the central values, which contrasts with the uniform distribution pattern of the true labels. This suggests that, in the absence of the Wide component, TPMN tends to make more conservative predictions, favoring scores within a middle range rather than covering the entire scoring spectrum. This predictive behavior aligns with TPMN's reduced ability to effectively leverage individual-specific information. It also highlights the role of the Wide component in retaining and utilizing personalized data, such as age and gender, which are crucial for generalized predictions of depression severity. In the ablation experiment, the exclusion of the Wide component suggests that TPMN is unable to capitalize on these essential personalized features, leading to decreased predictive accuracy and generalization capability.

TPMN, when including the Wide component, has shown improved stability and generalization capability across multiple experiments, without signs of overfitting, as depicted in Figs. 8 and 9(a). This further emphasizes the importance of the Wide component in enhancing TPMN's performance.

5.3.2. Ablation experiment of the prompting component

Removing the prompt component led to a significant increase in TPMN's MAE to 10.46, and a corresponding rise in RMSE to 14.24, as shown in Table 6. This outcome highlights the pivotal role of the prompting component in maintaining TPMN's predictive accuracy. As shown in Fig. 8, the NoPrompt variant shows higher MAE and RMSE values, with the validation curves plateauing early. This suggests that the model struggles to capture clinically relevant patterns without topic prompts, leading to reduced performance. The results underscore the importance of topic prompts in guiding the model to prioritize key features and adapt to individual differences.

As illustrated in Fig. 9(c), a deeper analysis of the score distribution revealed that, without the prompting component, TPMN's predicted scores exhibited a pronounced skew to the left side of the distribution, markedly deviating from the uniform distribution trend of the true labels. This indicates that, in the absence of prompting information, TPMN fails to capture the diversity and complexity inherent in the data, resulting in predictions that are disproportionately concentrated on certain values.

The change in distribution is linked to TPMN's over-reliance on specific features when the prompting component is absent. The prompting component provides additional contextual information, enabling TPMN to better understand the diversity of the data, and thus make more accurate predictions. It assists TPMN in learning more representative features during training, allowing the model to generalize better when confronted with unseen data.

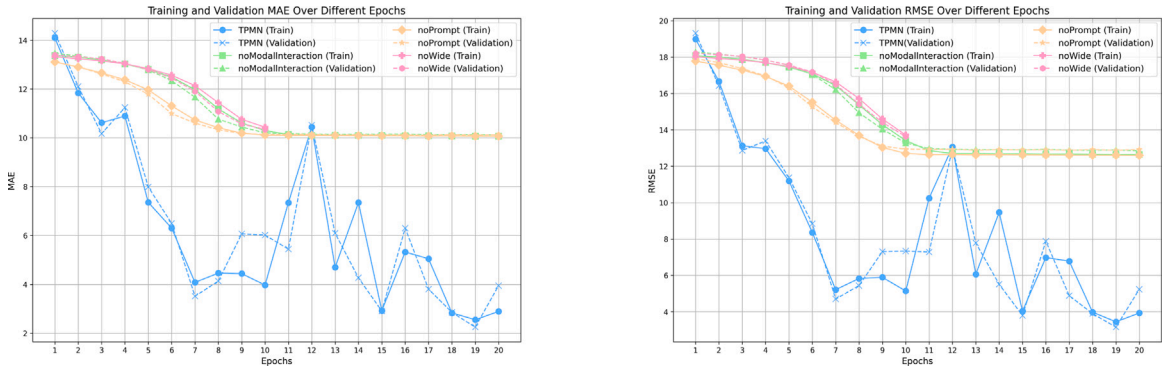

Fig. 8. Schematic diagram of changes in MAE and RMSE during the training process.

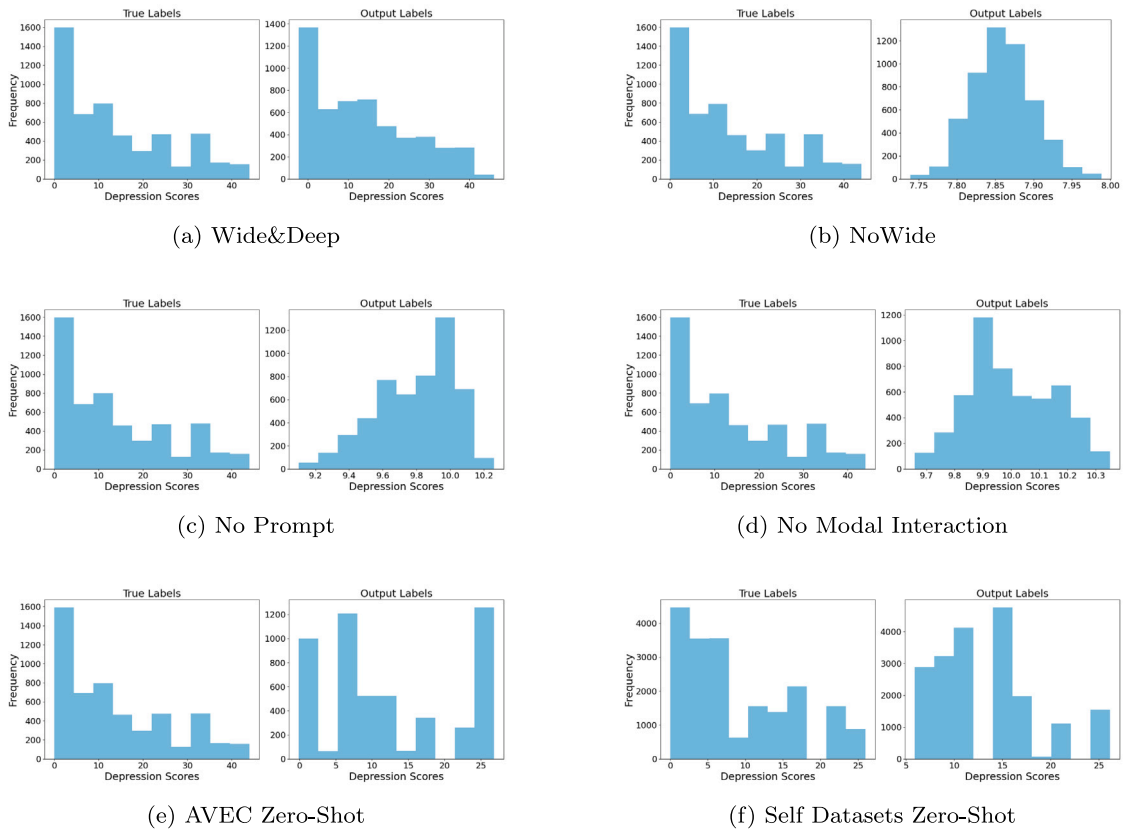

Fig. 9. Histogram of the distribution of predicted and true values during the experimental process.

5.3.3. Ablation experiment of coupling function

Upon the removal of the coupling functions, TPMN's MAE increased to 10.12, and the RMSE rose to 12.92, indicating a substantial degradation in performance compared to TPMN with the coupling functions intact, as shown in Table 6. Additionally, the distribution of predicted depression scores was skewed towards the left side of the actual score distribution, suggesting a tendency for the model to assign lower scores in the absence of coupling functions.

The NoModal Interaction variant performed worse than TPMN, showing slower convergence and higher final errors in both training and validation, as shown in Fig. 8. This is because the absence of coupling functions prevents effective cross-modal interaction, leading to incomplete integration of multimodal information. The observed phenomenon can be attributed to the limited effectiveness of directly using encoded text information in video and audio prompts, which hampers the ability of these prompts to influence the respective modalities. Furthermore, the impact of text prompts is minimal, given that TPMN uses BERT for text feature extraction, which can only capture limited information from interview topics.

Coupling functions are crucial for integrating information from different sources. Without them, the model fails to fully exploit the complementary information inherent in multimodal data, resulting in reduced predictive accuracy. These findings highlight the importance of coupling functions in multimodal depression recognition tasks.

5.3.4. Ablation experiments on different data subsets

Experiments were conducted on various data subsets (different ages, different genders) to assess the model's performance across demographic groups. The results are shown in Table 7. While the performance on male and female subsets is comparable, the female subset shows slightly better results, which may be attributed to a higher number of female samples in the training data or the model's better ability to capture gender-specific features. For example, male samples may

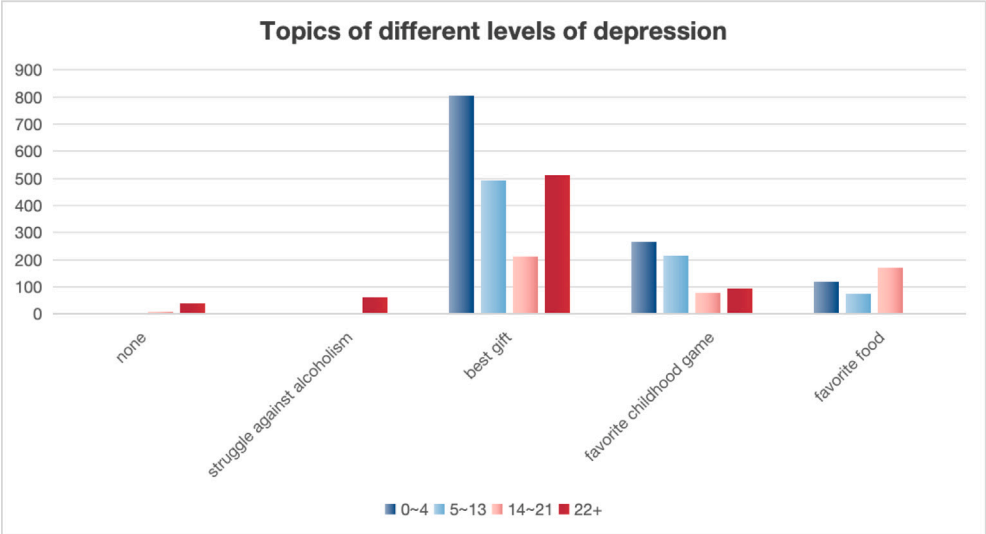

Fig. 10. Topics of different levels of depression.

Table 7
Data subsets experiment.

Subset category	Datasets	MAE	RMSE
Male	AVEC2014	5.13	6.61
Female	AVEC2014	5.12	6.36
0–30	AVEC2014	6.12	6.44
30–60	AVEC2014	6.12	7.01
60+	AVEC2014	7.27	7.08

exhibit more prominent facial hair, which can obscure facial expressions, whereas female samples may display richer micro-expressions. The model's performance on the 60+ age group is notably worse, likely due to the smaller number of samples in this age group within the data distribution. The model may struggle to adapt to age-specific features, such as the impact of wrinkles on facial recognition, age-related vocal tremors, changes in speech rate, and differences in language style. This paper also analyzes topic preferences across different depression levels (0–4, 5–13, 14–21, 22+), as shown in Fig. 10. The results highlight significant differences in the topics that individuals with varying depression levels are interested in.

For instance, the topic "best gift" is more prominently discussed among individuals with higher depression levels (22+), possibly indicating that people with more severe depression may focus more on this subject. On the other hand, topics like "favorite food" and "favorite childhood game" are more prominent among those with lower depression levels (0–4). This suggests that individuals with different levels of depression may engage differently with various topics, which could inform the development of personalized prompts tailored to specific depression levels. TPMN achieves the lowest MAE (2.70) and RMSE (3.65) on the AVEC2014 dataset, outperforming the ablation variants. This highlights the importance of integrating personal information (Wide part), topic prompts (Prompt part), and cross-modal interactions (Modal Interaction part) for robust depression severity prediction.

5.4. Our dataset experiment

To substantiate the effectiveness of TPMN, its performance against existing baseline methods on our self-collected dataset. We trained both baseline methods and TPMN on our dataset, and the resulting training curves are presented in Fig. 11. We selected CNN+LSTM and transformer baselines for comparison due to their widespread use and strong performance in various sequence modeling tasks. All models were trained for 10 epochs with similar hyperparameters to ensure a fair comparison. Comparison with Baselines: As depicted in Fig. 11, TPMN demonstrated superior performance over baseline methods. Notably, TPMN achieved lower MAE and RMSE values, indicating better predictive accuracy and stability. The TPMN demonstrated superior performance over baseline methods on our dataset with 43 participants, although its performance was lower compared to that on the AVEC2013 and AVEC2014 datasets. Consequently, we conducted additional experiments using different group compositions: 10 depressed individuals and 10 normal individuals, as well as 15 depressed individuals and 15 normal individuals. With an increased number of training samples (from 10 depressed and 10 normal individuals to 15 depressed and 15 normal individuals), TPMN's performance improved significantly. The improvement in performance with increased sample size suggests that TPMN effectively captures shared characteristics among individuals, leading to better feature representation and, consequently, improved predictive performance. The experiments on our dataset provide a compelling demonstration of TPMN's effectiveness. The observed performance improvements with increased training data underscore the model's capacity to leverage larger datasets for enhanced learning. These findings validate TPMN as a promising approach for depression recognition tasks, offering a solid foundation for future research and practical applications in mental health.

5.5. Zero-shot experiment

In zero-shot experiments, TPMN achieved an MAE of 6.42 and an RMSE of 8.9 on the AVEC2014 dataset, and an MAE of 7.96 and an RMSE of 8.79 on our self-collected dataset of Chinese participants, as shown in Table 8. These results are comparable to baseline studies, demonstrating that TPMN is competitive with current techniques. The performance gap is primarily attributed to cultural and linguistic differences between the German-speaking AVEC datasets and the Chinese dataset. For instance, cultural norms may influence the expression and perception of depression, leading to variations in emotional cues across datasets. Additionally, the smaller sample size of the Chinese dataset (43 participants) limits the model's ability to fully adapt to the new cultural context.

Fig. 9(e) and (f) show that in zero-shot experiments, the distribution of TPMN's predicted scores is broader and significantly closer to the actual distribution compared to the ablation experiments. This suggests that TPMN is able to capture the full spectrum of depression scores, including varying degrees of depressive tendencies. Despite cultural differences in language expression, emotional expression, and social

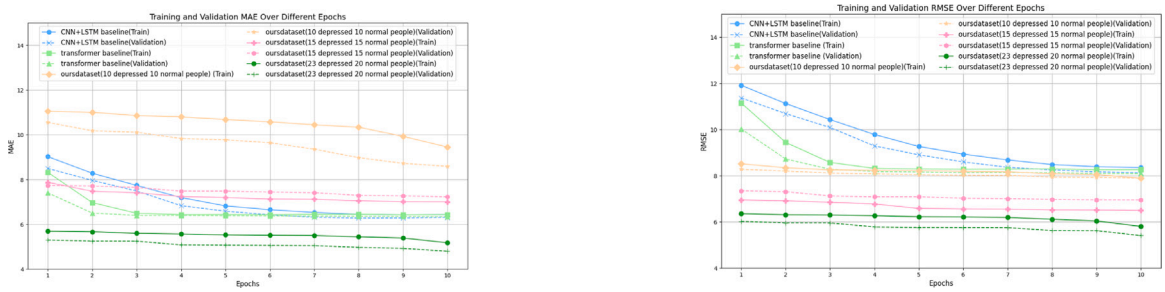

Fig. 11. Schematic representation of the changes in MAE and RMSE during the training process in the self-collected dataset.

Table 8
Zero-shot experiment.

Methods	Datasets	MAE	RMSE
TPMN(Train)	AVEC2013	3.01	3.07
TPMN (zero-shot)	AVEC2014	6.42	8.9
TPMN (zero-shot)	Our Dataset	7.96	8.79

behavioral norms, TPMN demonstrates stable performance across both datasets. This can be attributed to TPMN's multimodal feature extraction capabilities, which allow it to capture commonalities across cultures. The model's robust performance on datasets from two distinct cultural backgrounds suggests that TPMN has a certain level of cross-cultural generalizability. This finding is significant for the development of depression recognition tools that could be applied globally. In zero-shot experiments across cultural datasets, TPMN demonstrated relatively stable performance, with a slight decrease in MAE observed across different cultural backgrounds. This variation is attributed to cultural differences and distinct data distributions. Cultural disparities likely induce variations in both the expression and perception of depression across populations. For instance, some cultures may internalize depressive symptoms, while others may externalize them more readily. Additionally, different social and cultural contexts can influence how individuals understand and articulate depression, thereby affecting the score distributions. The score distribution in the Chinese dataset reflects the unique expressions and perceptions of depressive symptoms within the Chinese population, particularly in relation to the factors that contribute to depression, which may differ from those in the German population.

The distribution of depressive symptoms and the quality of annotations vary across datasets. The AVEC datasets, with their focused task design and high-quality annotations, offer more consistent signals for model training. In contrast, our self-collected dataset has a smaller sample size and potentially more variable annotations. For example, while the AVEC datasets may include topics such as "best gift" and "favorite food", our self-collected dataset centers on structured interview topics like "family" and "work". These differences make it more challenging to precisely match relevant topic prompts in zero-shot experiments, leading to higher MAE values. The discrepancies in topic focus between datasets are largely attributed to cultural differences. The AVEC datasets, primarily collected from German-speaking populations, reflect topics more relevant to that cultural context, whereas our self-collected dataset, gathered from a Chinese population, emphasizes topics that resonate more strongly within that cultural framework. Additionally, the limited sample size and demographic diversity in our self-collected dataset may constrain the model's ability to capture the full spectrum of depressive symptoms. In contrast, the larger and more diverse AVEC datasets enable the model to learn more robust patterns.

6. Conclusion

The study has developed and validated a generalized multimodal depression severity prediction model that leverages topic prompts. Experimental evaluations on the AVEC2013/2014 datasets and a self-collected Chinese dataset demonstrate TPMN's excellent performance and robust generalization capabilities. The personalized module, prompting module, and coupling function module are key to enhancing predictive accuracy. Its performance across cultural datasets confirms its broad applicability. This approach represents a significant advancement and has the potential to stimulate further research in disease recognition.

Future work will focus on optimizing the personalized module, prompting module, and intermodal interaction module to better capture individual variations. Expanding training to include datasets from a wider range of cultural backgrounds will further improve the model's generalization capabilities. Enhancing the quality and consistency of annotations in the self-collected dataset will reduce noise and provide a stronger learning signal for the model. Additionally, we will explore cross-cultural adaptation techniques, such as domain adaptation and transfer learning, to boost the model's performance on datasets from diverse cultural contexts. For example, fine-tuning the model on a small amount of target cultural data could help bridge the performance gap.

CRediT authorship contribution statement

Di Zhao: Writing – original draft, Investigation, Writing – review & editing, Methodology, Software. **Qi Li:** Supervision, Writing – review & editing, Methodology. **Qingyu Na:** Software, Writing – original draft, Writing – review & editing, Methodology. **Hongyu Xie:** Data curation, Investigation. **Tingyun Jiang:** Data curation, Writing – review & editing.

Declaration of Generative AI and AI-assisted technologies in the writing process

The authors of this scientific paper have utilized generative artificial intelligence (AI) tools in the writing process. Specifically, we have employed AI-powered language models to assist with drafting sections of the manuscript, generating ideas, and refining the overall structure and clarity of the text. The AI has been instrumental in providing suggestions for improving the coherence and readability of our work, as well as offering alternative phrasings and terminology that enhance the precision and impact of our scientific communication. However, it is important to note that the use of generative AI has been carefully managed to ensure the integrity and originality of our research. All AI-generated content has been thoroughly reviewed, edited, and validated by the authors. We have made certain that the AI's contributions align with our research objectives, data, and findings, and that they do not compromise the accuracy or authenticity of our scientific work. The authors have maintained full control over the final content and conclusions presented in this paper. We acknowledge the potential benefits of using generative AI in scientific writing, such as increased efficiency and improved quality of written work. However, we also recognize the importance of human oversight and expertise in ensuring the validity and reliability of scientific research. Therefore, we have utilized AI as

a complementary tool to support our writing process, while upholding the highest standards of scientific rigor and integrity. The authors take full responsibility for the content and conclusions of this paper, and any errors or inaccuracies that may be present are solely attributable to us. The use of generative AI has been disclosed transparently to maintain transparency and trust in the scientific community.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Qi Li reports financial support was provided by the Science and Technology Development Plan Project of Jilin Province, China (20240101344JC, 20230203098SF). Qi Li reports financial support was provided by the Zhongshan Social Welfare and Basic Research Project, China (2023B2015). If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.inffus.2025.103554.

Data availability

The data presented in this study were collected through a combination of structured interviews and self-administered depression scales. All participants were patients diagnosed with depression at The Third People's Hospital of Zhongshan, Zhongshan, Guangdong, China (Zhongshan Mental Health and Psychiatric Prevention and Treatment Center), as per the International Classification of Diseases, 11th Revision (ICD-11) criteria. The data collection process involved obtaining participants' personal information, quantitative data on depressive symptoms from the depression scales, and qualitative insights from structured interviews about their experiences with depression, treatment feelings, and expectations for future treatment. The study aimed for a target sample size of 50 patients, with 23 participants currently enrolled, selected based on both clinical diagnosis and depression scale scores. The raw interview data, including audio recordings and transcripts, as well as the completed depression scales, are stored securely at The Third People's Hospital of Zhongshan. Due to the sensitive nature of the data, which includes personal health information and potentially identifying details, the data cannot be made publicly available. However, interested researchers may request access to anonymized data for the purpose of verification and further research, subject to approval by the hospital's ethics committee and compliance with relevant data protection regulations. Requests for data access should be directed to the corresponding author, who will facilitate the process in accordance with the hospital's policies and ethical guidelines. We confirm that all data collection and analysis methods were carried out in accordance with the approved research design and ethical protocols. The study was approved by the ethics committee (Approval SSYLL-KY-20240906), and informed consent was obtained from all participants or their legal guardians where applicable.

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